

Vulnerability Assessment Report

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System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 2025 to August 2025. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

Purpose

The database server is a critical asset to the business, as it stores and manages essential operational and customer data that support daily functions and decision-making. Securing the data on this server is vital to protect sensitive information from unauthorized access, data breaches, or loss. A successful attack or failure could lead to significant downtime, reputational damage, financial loss, and non-compliance with regulatory standards. This vulnerability assessment aims to evaluate and strengthen access controls to mitigate such risks and ensure the system's integrity, availability, and confidentiality.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Competitor	Disrupt mission-critical operations.	2	3	6
Supplier	Craft counterfeit certificates	1	3	3

	<i>cates.</i>			
<i>Hacker</i>	<i>Install persistent and targeted network sniffers on organizational information systems.</i>	3	3	9

Approach

The selected threat sources—competitors, suppliers, and hackers—represent significant and realistic risks to the organization. Competitors may attempt to disrupt mission-critical operations to gain a strategic advantage, which could impact business continuity and customer trust. Suppliers pose a risk by potentially introducing counterfeit certificates, undermining the integrity of secure communications. Hackers are a high-priority threat due to their ability to install persistent network sniffers that can silently extract sensitive data. These threats were chosen based on their relevance, potential impact, and likelihood in the current business and technological environment.

Remediation Strategy

To mitigate the identified risks, the organization should implement the **principle of least privilege** to ensure users and systems only have access necessary for their roles, reducing the impact of compromised accounts. **Defense in depth** should be applied by layering security measures such as firewalls, intrusion detection systems, and endpoint protection to defend against persistent threats like network sniffers. **Multi-factor authentication (MFA)** can strengthen user verification and help prevent unauthorized access. Finally, using the **Authentication, Authorization, and Accounting (AAA)** framework ensures robust identity management, access control, and monitoring of user activities across the system.